Verified Data Sources for Extended-Range Sand and Dust Storm (SDS) AI Forecasting

Data Acquisition and Source Verification Document

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1 Purpose

This document provides a consolidated list of official and reliable data sources relevant to the research project on Extended-Range Sand and Dust Storm (SDS) AI Forecasting and Socioeconomic Analytics.

The sources cover the major data modalities required for the proposed forecasting framework:

1. Numerical Weather Prediction (NWP)
2. Atmospheric Reanalysis
3. Satellite Earth Observation
4. Ground-Based Observations
5. Air Quality and Environmental Monitoring
6. Sub-seasonal-to-Seasonal (S2S) Climate Indices
7. GIS, DEM, and Land-Cover Data
8. Socioeconomic and Public Survey Data

The links below prioritize official institutional portals such as ECMWF, Copernicus, NASA, NOAA, CMA, NSMC, USGS, and ESA.

2 1. Numerical Weather Prediction (NWP)

NWP data provide atmospheric forecasts that can be used as dynamic meteorological predictors for SDS forecasting.

Important variables include:

- 10-m zonal wind (u_{10})
- 10-m meridional wind (v_{10})
- Mean sea-level pressure (MSLP)

- 2-m air temperature
- Boundary-layer height (BLH/PBLH)
- Geopotential height
- Vertical wind shear
- Atmospheric circulation variables

2.1 ECMWF Open Data

Provider: European Centre for Medium-Range Weather Forecasts (ECMWF)

Official source:

ECMWF Open Data

`https://www.ecmwf.int/en/forecasts/datasets/open-data`

Use in this research:

- Operational weather forecasts
- Wind fields
- Pressure fields
- Temperature
- Atmospheric circulation
- Forecast lead times for extended-range prediction

Important note: The currently documented public ECMWF Open Data should not automatically be described as providing the project's proposed 0.125° resolution. Higher-resolution or archived products may require different access arrangements.

3 2. Atmospheric Reanalysis

Reanalysis datasets provide historically consistent atmospheric states and are useful for model training, climatology, hindcasts, and validation.

3.1 Copernicus Climate Data Store

Provider: Copernicus Climate Change Service (C3S)

Official portal:

Copernicus Climate Data Store

`https://cds.climate.copernicus.eu/en/`

3.2 ERA5 Complete Global Reanalysis

Official dataset:

ERA5 Complete Global Reanalysis

`https://cds.climate.copernicus.eu/datasets/reanalysis-era5-complete`

Potential variables:

- 10-m wind components
- 2-m temperature
- Mean sea-level pressure
- Geopotential
- Boundary-layer variables
- Soil moisture
- Surface pressure
- Atmospheric pressure-level variables

Use in this research:

ERA5 can provide historical atmospheric conditions for training, validation, climatological analysis, and comparison with operational forecast data.

4 3. Satellite Earth Observation

Satellite observations provide spatial information about dust plumes, aerosols, land surface conditions, vegetation, soil moisture, and temperature.

4.1 3.1 NASA Earthdata

Official portal:

NASA Earthdata

`https://www.earthdata.nasa.gov/`

NASA Earthdata provides access to a large collection of Earth observation datasets.

4.2 3.2 MODIS and MAIAC

Official NASA collection search:

NASA LP DAAC MODIS/MAIAC Collections

`https://cmr.earthdata.nasa.gov/search/site/collections/directory/LPCLOUD/gov.nasa.eosd`

Relevant products may include:

- MODIS Terra
- MODIS Aqua
- Aerosol Optical Depth (AOD)
- MAIAC products
- Vegetation information
- Land-surface information

Example product: MCD19A2 MAIAC aerosol optical depth.

4.3 3.3 NASA Worldview

Official source:

NASA Worldview

`https://worldview.earthdata.nasa.gov/`

Worldview can be used for visual inspection of satellite observations and dust events.

4.4 3.4 FengYun Satellite Data

Provider: China Meteorological Administration / National Satellite Meteorological Center (NSMC)

Official FengYun data portal:

NSMC FengYun Data Portal

`https://data.nsmc.org.cn/DataPortal/en/home/index.html`

4.5 3.5 FY-4B

Official source:

NSMC FY-4B Application

`https://www.nsmc.org.cn/nsmc/en/theme/FY4B_application.html`

Potential applications include:

- Dust plume monitoring
- Dust RGB products
- High-frequency geostationary observations
- Cloud and atmospheric monitoring

5 4. Ground-Based Meteorological Observations

Ground observations are important for creating ground-truth labels and validating satellite and model predictions.

Relevant measurements include:

- PM10
- PM2.5
- Visibility
- Wind speed
- Wind direction
- Temperature
- Pressure
- Relative humidity
- Weather phenomena and dust codes

5.1 4.1 China Meteorological Administration

Official data portal:

CMA Data Portal

`https://k.data.cma.cn/`

5.2 4.2 CMA Surface Meteorological Observations

Official dataset page:

CMA Surface Meteorological Observations

`https://k.data.cma.cn/mekb/?dataCode=A.0012.0001.S011&r=data/detail`

These observations can be used to construct meteorological predictors and to validate SDS events against ground observations.

6 5. Air Quality and Environmental Monitoring

Ground-based air-quality monitoring is particularly important for validating aerosol and particulate-matter predictions.

6.1 5.1 China National Environmental Monitoring Centre

Official website:

China National Environmental Monitoring Centre (CNEMC)

`https://www.cnemc.cn/`

6.2 5.2 CNEMC Air Quality Platform

Official platform:

CNEMC Air Quality Monitoring Platform

`https://air.cnemc.cn/`

Potential variables include:

- PM10
- PM2.5
- AQI
- Other air-quality indicators

Historical bulk-data availability should be verified according to the specific access mechanism and period required by the research.

7 6. Sub-seasonal-to-Seasonal Climate Indices

Large-scale atmospheric circulation and teleconnection indices can provide additional predictors for extended-range SDS forecasting.

Relevant indices include:

- Arctic Oscillation (AO)
- North Atlantic Oscillation (NAO)
- El Niño–Southern Oscillation (ENSO)
- Siberian High-related circulation indicators

7.1 6.1 NOAA Climate Prediction Center

Official source:

NOAA Climate Prediction Center

`https://www.cpc.ncep.noaa.gov/`

7.2 6.2 NOAA Teleconnection Indices

Official source:

NOAA Teleconnection Patterns

`https://www.cpc.ncep.noaa.gov/products/precip/CWlink/daily_ao_index/teleconnections.sh`

7.3 6.3 NOAA Daily Arctic Oscillation

Official source:

NOAA Daily Arctic Oscillation Index

`https://www.cpc.ncep.noaa.gov/products/precip/CWlink/daily_ao_index/new.ao.shtml`

These indices can be incorporated as temporal or lagged predictors for extended-range forecasting.

8 7. GIS, DEM, and Land-Surface Data

Geospatial datasets provide relatively static or slowly varying information about the physical environment.

Potential features include:

- Elevation
- Slope
- Aspect
- Land cover
- Desert extent
- Soil properties
- Dust-source regions
- Transport corridors

8.1 7.1 NASA LP DAAC

Official source:

NASA Land Processes Distributed Active Archive Center (LP DAAC)

`https://www.earthdata.nasa.gov/centers/lp-daac`

Relevant datasets include elevation products such as SRTM-derived products.

8.2 7.2 USGS Landsat

Official Landsat Data Access:

USGS Landsat Data Access

`https://www.usgs.gov/landsat-missions/landsat-data-access`

Official Landsat Data and Products:

USGS Landsat Data and Products

`https://www.usgs.gov/landsat-missions/data`

Landsat can provide high-resolution information for land-surface and environmental characterization.

8.3 7.3 ESA WorldCover

Official data access:

ESA WorldCover Data Access

`https://esa-worldcover.org/en/data-access`

WorldCover can be used for land-cover classification and environmental feature extraction.

Important note: The documented WorldCover maps should not be assumed to provide a new annual global map for every year from 2020 onward. The specific available product year should be checked before dataset construction.

9 8. Data Source Summary

Data Type	Provider	Official Source
NWP Forecasts	ECMWF	ECMWF Open Data
Reanalysis	Copernicus	Copernicus Climate Data Store
ERA5	Copernicus	ERA5 Complete Global Reanalysis
MODIS / MAIAC	NASA	NASA Earthdata
FengYun	NSMC/CMA	NSMC FengYun Data Portal
FY-4B	NSMC/CMA	FY-4B Official Page
Meteorological Stations	CMA	CMA Data Portal
Air Quality	CNEMC	CNEMC Air Quality Platform
Climate Indices	NOAA CPC	NOAA Climate Prediction Center
Teleconnections	NOAA CPC	NOAA Teleconnections
Elevation	NASA LP DAAC	NASA LP DAAC
Landsat	USGS	USGS Landsat Data
Land Cover	ESA	ESA WorldCover

10 9. Source Verification and Reliability

The listed sources were checked against official institutional portals. The verification focused on confirming that each link corresponds to the responsible organization or its official data distribution platform.

The main source organizations are:

- ECMWF
- Copernicus Climate Change Service
- NASA

- National Satellite Meteorological Center (NSMC)
- China Meteorological Administration (CMA)
- China National Environmental Monitoring Centre (CNEMC)
- NOAA
- USGS
- European Space Agency (ESA)

These sources provide an appropriate foundation for building a reliable SDS forecasting dataset.

However, source verification does not automatically guarantee that every dataset is freely downloadable, unrestricted, or available at the exact temporal and spatial resolution required by the research. Registration, licensing, API access, archive access, and product-specific restrictions must be checked during the acquisition stage.

11 10. Important Data-Resolution Considerations

The proposed research may specify particular spatial and temporal resolutions for different datasets. These specifications should be treated as project requirements rather than assumptions about the public availability of every source.

For example:

- ECMWF public Open Data has specific documented resolution and availability conditions.
- ERA5 provides a long historical reanalysis record but has a different native resolution from some proposed modeling grids.
- Satellite products have different spatial and temporal resolutions depending on the sensor and product.
- Ground stations provide high temporal resolution but are spatially irregular.
- Land-cover products may only be available for particular reference years.

Therefore, preprocessing should include spatial and temporal harmonization before the datasets are supplied to the AI model.

12 11. Recommended Data Integration

The final SDS forecasting dataset can be constructed by integrating the following information:

$$NWP + ERA5 + Satellite + GroundStations + ClimateIndices + GIS$$

These data can subsequently be transformed into a common spatial grid and time axis.

A conceptual forecasting pipeline is:

$$DataAcquisition \rightarrow QualityControl \rightarrow Spatial/TemporalAlignment \rightarrow FeatureEngineering \rightarrow AI$$

13 12. Final Checklist

Before downloading the datasets, verify:

1. The source is an official institutional provider.
2. The dataset covers the required geographic domain.
3. The dataset covers the required historical period.
4. The temporal resolution is appropriate.
5. The spatial resolution is documented.
6. Required variables are available.
7. Data licensing and access conditions are understood.
8. Missing values and quality flags are documented.
9. Coordinate reference systems are documented.
10. Units are documented and standardized.
11. The dataset can be aligned with the other modalities.
12. The final dataset can be independently reproduced.

Conclusion

The sources in this document provide official starting points for the major data modalities required by the Extended-Range Sand and Dust Storm AI Forecasting research. The actual acquisition stage should still verify dataset version, temporal coverage, spatial resolution, access conditions, and variable availability for the exact study domain and period.